Shahjalal University of Science and Technology
Course No: EEE 339 (Power Electronics)
Term Test-1
Marks:20, Time:30 minutes

1. Name five of the processes to turn-on a thyristor. Expain the effect dv/dt in turning-on of a tyristor with diagram (5)
2. Tabulate the similarity and dissimilarity between Power BJT and IGBT in terms of power handling capacity, switching speed, on-state resistance and blocking voltage. (4)
3. Show the Load Voltage v_o,current i_A, voltage across DIAC and TRIAC with proper alignment for the diagram shown in Figure-1 (3)
4. A sinusoidal voltage source of $v(t) = 2 + 5\ \cos(2t) + 3\ \cos(2 \times 2t + 45°)\ V$ is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as

$$i(t) = 1.5 + 2\ \cos(2t + 20°) + 1.1 \cos(2 \times 2t - 20°)\ A$$

Determine (*a*) the power absorbed by the load, (*b*) the power factor of the load, (*c*) the distortion factor of the load current, (*d*) the total harmonic distortion of the load current. (8)

Ex 2-19

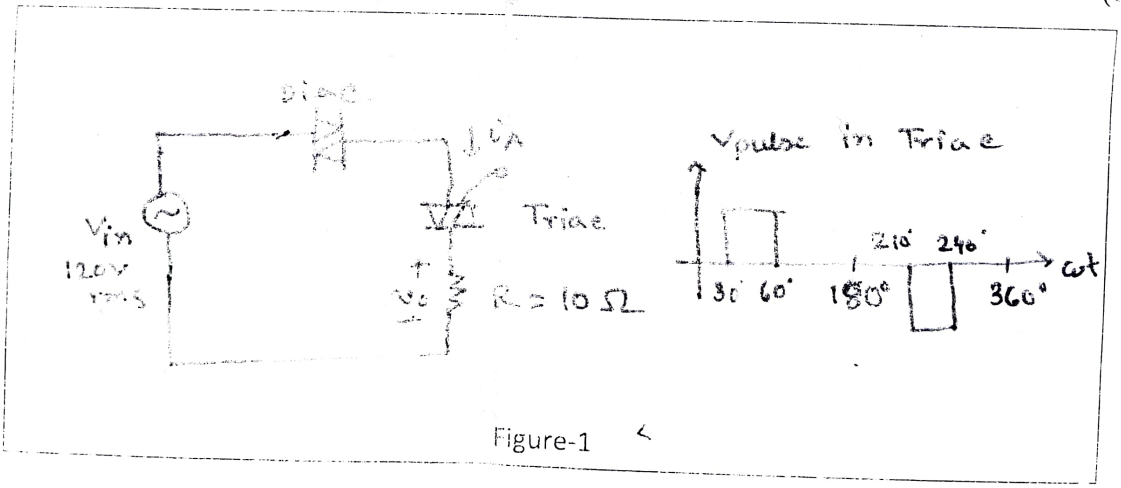


Figure-1

Shahjalal University of Science and Technology
Course No: EEE 339 (Power Electronics)
Term Test-2
Marks:20, Time:30 minutes

1. Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and capacitor current for a typical buck-boost converter. (5)

2. A boost converter has parameters, Vs=20 V, D=0.6, R=12.5 Ω, L=10 μH and C=40 μF. The switching frequency is 200 kHz. a) Determine the output voltage, b) Determine the average, minimum and maximum inductor current c) The output ripple voltage, d) the average current in the diode. (7.5)

Ex 6-17

3. Draw the switching diagram for fifth harmonic elimination in a full-bridge inverter output voltage control scheme. (2.5)

4. For a full bridge inverter, the dc source is 125 V, the RL load value is 10 Ω and 20 mH respectively. The fundamental switching frequency is 60 Hz. Determine the value of switching angle α to obtain output amplitude of 90 V at the fundamental frequency. (5)

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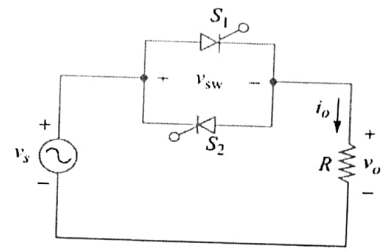
Department of Electrical & Electronic Engineering
3rd Year 2nd Semester Final Examination – 2021
Session: 2018-19
Course No: EEE 339
Course Title: Power Electronics
Credits : 3.0 Full Marks : 100 Time : 3 Hours

[Answer any four questions taking two from each part]

PART A

Q1. a) Differentiate among Power BJT, MOSFET and IGBT in terms of power handling capacity, input impedance, switching speed, on-state resistance and blocking voltage. 7

b) With proper alignment, draw the waveform of the output voltage v_o, output current i_o and voltage across the switches v_{sw}, for the following circuit when delay angle α=15° and extinction angle,β=120°. 6



c) A nonsinusoidal periodic voltage has a Fourier series of $v(t) = 10 + 8\cos(2\pi 60t - 25°) + 6\cos(4\pi 60t + 20°)\ V$ which is connected to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as

$$i(t) = 3 + 4\cos(2\pi 60t + 30°) + 1.5\cos(4\pi 60t + 45°)\ A$$

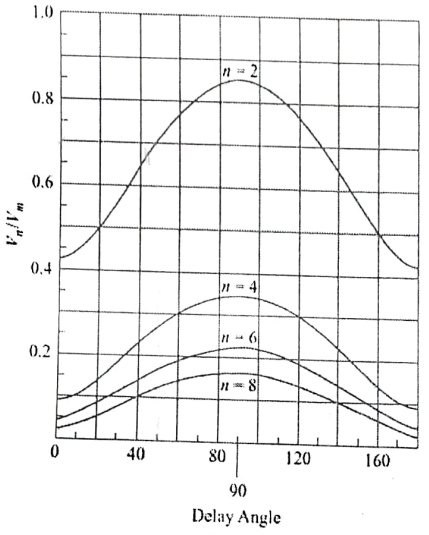
Determine (i) rms value of the current signal (ii) the power absorbed by the load, (iii) the power factor of the load, (iv) the distortion factor of the load current, (v) the total harmonic distortion of the load current, (vi) the crest factor of the load current. 12

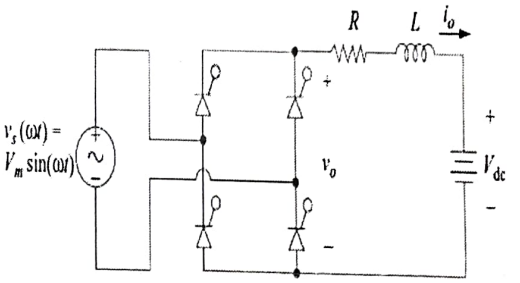
Q2. a) Discuss the significance of Reverse recovery time of a diode. 5

b) Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model . 8

c) A half-wave rectifier has a 120 V rms 60 Hz ac source. The load is 50 W. Determine the value of a filter capacitor to keep the peak-to-peak ripple across the load to less than 1.5 V. Also determine the average and peak values of diode current. 6

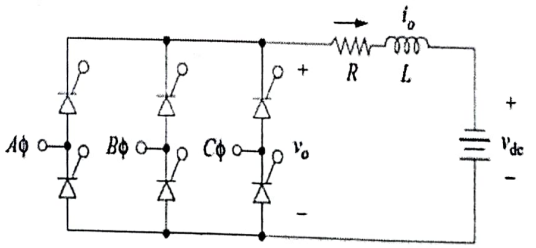
d) The controlled full-wave rectifier of Figure below has an ac source of 240 V rms at 60 Hz, V_{dc} =100 V, R =5 Ω, and an inductor large enough to cause continuous current. (*i*) Determine the delay angle α such that the power absorbed by the dc source is 1000 W. (*ii*) Determine the value of inductance that will limit the peak-to-peak load current variation to 2 A. You may use the delay angle vs harmonics curve to solve the problem by finding impedance for the 2nd harmonics. 6





Q3 a) Write short note on single phase cycloconverter. 5

b) A three-phase converter is operating as an inverter and is connected to a 300-V dc source as shown in Fig. The ac source is 240 V rms line to line at 60 Hz. The resistance is 0.5 Ω, and the inductor is large enough to consider the load current to be ripple-free. (*a*) Determine the delay angle α such that the output voltage of the converter is Vo = - 280 V. Determine the power supplied or absorbed by each component in the circuit. The SCRs are assumed to be ideal. 10



c) Briefly describe any one kind of thyristor forced commutation circuit with necessary diagram and waveforms. 6

d) With diagram, explain the pinch-off condition in a Power MOSFET. 4

PART B

Q4 a) Draw the circuit diagram for a typical boost converter. Draw the equivalent circuits for switch-on and off conditions. With proper alignment, draw the waveform of inductor current, load current, output voltage, capacitor current for the circuit. Derive the relation between input and output voltage. Also derive the relation between inductor current and diode current from the waveforms. 10

b) The typical buck-boost converter has V_s = 24 V, V_o=- 36 V, and a load resistance of 10 Ω. If the switching frequency is 100 kHz, (a) determine the inductance such that the minimum current is 40 percent of the average and (b) determine the capacitance required to limit the output voltage ripple to 0.5 percent. 6

Ex 6-25

c) The manufacturer of a selected diode gives the rate of fall of the diode current $^{di}/_{dt} = 20\ A/\mu s$, and its reverse recovery time $t_{rr} = 5\ \mu s$. What value of peak reverse current do you expect? 4

d) Draw the circuit of a SEPIC converter. What is the advantage of it over Buck-Boost converter? 5

Q5 a) Briefly explain the types of switch mode DC-DC converters used in a standalone solar PV system. 5

b) Draw the circuit of a four-source cascade multilevel converter that features the equalization of source power. Show the voltage outputs across each H bridge and the total output pattern. 8

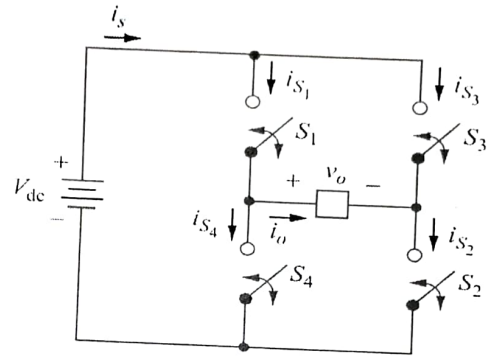
c) The full-bridge inverter is used to produce a 60-Hz voltage across a series *RL* load using bipolar PWM. The dc input to the bridge is 100 V, the amplitude modulation ratio m_a is 0.8, and the frequency modulation ratio m_f is 21. The load has a resistance of R = 10 Ω and series inductance L = 20 mH. Determine (*i*) the amplitude of the 60-Hz component of the output voltage and load current, (*ii*) the power absorbed by the load resistor, and (*iii*) the THD of the load current. Use the table shown below. 12

Exp 8-8 (Hart)

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	m_a=1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n=1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n=mf\pm2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

Q6 a) The full-bridge inverter in figure below has a switching sequence that produces a square wave voltage across a series RL load. The switching frequency is 60 Hz, V_{dc}=100 V, R = 10 Ω and L = 25 mH. Determine (i) the expression for load current, (ii) the power absorbed by the load resistor, and (iii) the average current in the dc source. 10



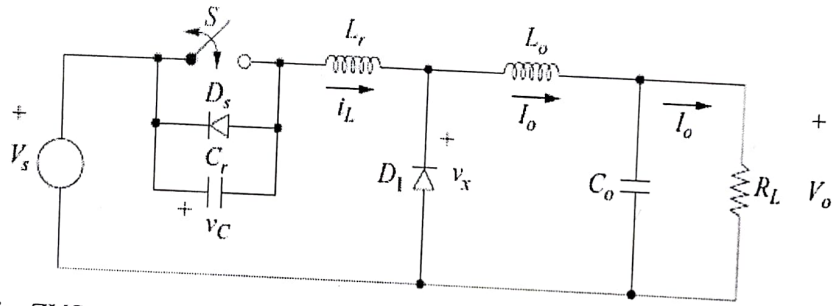
Exp 8-1

b) Draw the switching diagram for a square wave inverter that may reduce the third harmonics from the output. What is the amplitude of fundamental frequency for this switching pattern? 6

c) Show the diagram of a single phase dual converter drive circuit to control the speed of a separately excited dc-motor.

d) For the following ZVS resonant converter, Vs =5 V, Io =3 A, Lr =1 μH, and Cr =0.01 μF. 3

(i) Determine the output voltage when fs = 500 kHz. (ii) Determine the switching frequency such that the output voltage is 2.5 V. 6



Exp 9-10

Equations for ZVS resonant converter:

$$t_1 = \frac{V_s C_r}{I_o} \qquad \omega_0 = \frac{1}{\sqrt{L_r C_r}} \qquad Z_0 = \sqrt{\frac{L_r}{C_r}} \qquad V_o = V_s\left[1 - f_s\left(t_3 - \frac{t_1}{2}\right)\right]$$

$$t_2 = \frac{1}{\omega_0}\left[\sin^{-1}\left(\frac{-V_s}{I_o Z_0}\right)\right] + t_1 \qquad t_3 = \left(\frac{L_r I_o}{V_s}\right)\{1 - \cos[\omega_0(t_2 - t_1)]\} + t_2$$

Shahjalal University of Science and Technology
Department of Electrical and Electronic Engineering
3rd Year 2nd Semester Final Examination, 2022
Course Code: EEE-339, Session: 2019-20
Course Title: Power Electronics

Full Marks: 100 | **Credit: 3.0** | **Exam Duration: 3 Hours**

Part – A

*(Answer any **two** of the following questions)*

1. a. Write short notes on the following topics. 8
   i. Total Harmonic Distortion
   ii. Freewheeling Diode
   b. Derive the equation for average current, rms current and power factor for a half wave diode rectifier on RL-source load. 8
   c. For a half wave diode rectifier R = 25 Ω and L = 150 mH. The AC source is 120 V rms at 60 Hz. Determine (a) an expression for the current in the circuit, average current, rms current, the power absorbed by the RL load, and the power factor. 9

2. a. Derive the equation for output voltage, load current, minimum inductance and voltage ripple for a Buck-Boost converter. 8
   b. Explain how the capacitor resistance and inductor resistance affects the DC-DC converter output. 7
   c. Design a buck converter that has an input voltage of 5 V and an output of 3.3 V. The output current varies between 1 A to 2 A. The output voltage ripple must not exceed 2 percent. Chose an inductor such that the peak-to-peak variation in inductor current does not exceed 25 percent of the average value. Determine the required rms current rating of the inductor and maximum equivalent series resistance of the capacitor. 10

3. a. What are cycloconverter? What are they used for? 5
   b. Describe the working principle of a single-phase cycloconverter. 7
   c. Draw and describe the power electronics circuit based speed control method for induction motor. 7
   d. A parallel resonant DC-DC converter has V_s = 100 V, L_r = 8 μH, C_r = 0.32 μF, R_L =10 Ω, and f_s = 120 kHz. Determine the output voltage of the converter. Assume the output filter components L_o and C_o produce a ripple-free output current and voltage. 6

Part – B

*(Answer any **two** of the following questions)*

4. a. Compare between BJT and MOSFET as a switch. 5
   b. Explain the switching characteristics of a BJT. 6
   c. Using the two-transistor model explain the anode current in a thyristor or SCR. 6
   d. Describe the methods of a thyristor or SCR turn-on and turn-off. 8

5. a. Briefly explain AC voltage controllers. 3
   b. With proper diagram explain the principles of a square-wave inverter. 8
   c. The full-bridge inverter has a switching sequence that produces a square wave voltage across a series RL load. The switching frequency is 60 Hz, V_{dc} = 100 V, R = 10 Ω, and L = 25 mH. Determine an expression for load current, power absorbed by the load and average current in the DC source. 7
   d. How can we control amplitude and harmonics in an inverter? 7

6. a. For a controlled full wave bridge rectifier with a RL load describe the conditions for continuous and discontinuous current. 8
   b. Write the advantages of using IGBT as a switch. 4
   c. A full wave controlled bridge rectifier has an AC input of 120 V rms at 60 Hz and a 20 Ω resistive load. If the delay angle is 40 degrees, determine the average current in the load, the power absorbed by the load and the power factor. 6
   d. Briefly describe the half-cell model of an IGBT with appropriate figure. 7

Shahjalal University of Science and Technology
Course No: EEE 339 (Power Electronics)
Term Test-1
Marks:20, Time:30 minutes

1. Explain the reverse recovery time of diode with diagram. (4)
2. Draw the I-V curve of thyristor. Define Holding current and Latching current for thyristor (5)
3. With proper alignment, show the waveshape of output voltage, output current, diode currents and source current for a full wave bridge rectifier with sinusoidal source and R-L load. (5)
4. A sinusoidal voltage source of $v(t) = 240\sqrt{2}\sin(2\pi 60t)\ V$ is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as

$$i(t) = 8\ \sin(2\pi 60t) + 4\ \sin(4\pi 60t)\ A$$

Determine (*a*) the power absorbed by the load, (*b*) the power factor of the load, (*c*) the distortion factor of the load current, (*d*) the total harmonic distortion of the load current, (e) crest factor of the load current. (6)

type 2-10

Shahjalal University of Science and Technology
Course No: EEE 339 (Power Electronics)
Term Test-2
Marks:20, Time:30 minutes

1. Draw the circuit diagram, equivalent circuit for switch on and off condition and the waveforms of the inductor current, diode current and load voltage for a typical boost converter. (5)
2. A buck converter has parameters, Vs=50 V, D=0.4, R=20 Ω, L=400 μH and C=100 μF. The switching frequency is 20 kHz. a) Determine the output voltage, b) Determine the average , minimum and maximum inductor current c) The output ripple voltage. (7.5)
3. Draw the circuit diagram to control the speed of a separately excited dc motor by chopping field current supply. Explain the operation using required equations. (2.5)
4. A three phase rectifier is supplied by 480 V rms line to line 60 Hz source. The RL load is 100 Ω resistor in series with 15 mH inductor. Determine the rms load current and rms diode currents. (5)

Ex 4-36

Shahjalal University of Science & Technology
Department of Electrical & Electronic Engineering
3rd Year 2nd Semester Final Examination – 2023
Course No: EEE 339 Session: 2020-21
Course Title: Power Electronics

Credits : 3.0 **Full Marks : 100** **Time : 3 Hours**

[Answer any four questions taking two from each part]

PART A

Q1. a) Define distortion factor (DF) and total harmonic distortion (THD). Derive that

$$DF = \sqrt{\frac{1}{1 + (THD)^2}}$$ 7

b) Define reverse recovery current and reverse recovery time for diode switching. What is the importance of these parameters? 6

c) The manufacturer of a selected diode gives the rate of fall of the diode current di/dt=20 A/μs, and its reserve recovery time t_{rr}=5 μs. What value of peak reverse current and the storage charge do you expect? 6

d) A sinusoidal current of 10A rms at a 60 Hz fundamental frequency is contaminated with a ninth harmonic current. The current is expressed as 6

$$i(t) = 10\sqrt{2}\sin(2\pi 60t) + I_9\sqrt{2}\sin(18\pi 60t) \ A.$$

Determine the value of the ninth harmonic rms current I9 if the THD is –i) 5 percent ii) 20 percent.

Q2. a) Draw the two transistors model of a thyristor and explain the turn-on condition of a thyristor with equations from this model. 7

b) For the circuit of Fig. 2(b), the source is 120 V rms at 60 Hz, R=20 Ohm, L=0.04 H, and the delay angle is 45°. Determine (i) an expression for i(ωt) , (ii) the average current, (iii) the power absorbed by the load, and (iv) the power factor. 10

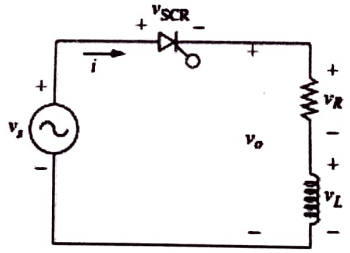


Fig 2(b)

c) A full-wave controlled bridge rectifier has an ac input of 120 V rms at 60 Hz and a 20 Ω load resistor. The delay angle is 40°. Determine the average current of the load, the power absorbed by the load, the apparent power of the sourced and the input pf. 8

Q3. a) Draw the configuration of three phase uncontrolled rectifier. Show the source voltages, output voltage, diode currents and line currents with proper alignment. 8

b) Draw the cross section of an IGBT showing the layers. Differentiate between punch-through and non-punch through IGBTs. 7

c) Write down the advantages of DC power transmission. 4

d) A three phase controlled rectifier has an input voltage of 480 V rms at 60 Hz. The load of the rectifier is resistor of 10 Ω in series with 50 mH inductance.

i) Determine the delay angle required to produce an average current of 50A in the load

ii) Determine the amplitude of the harmonics n=6 and n=12. For harmonics calculation, Fig. 3(d) can be utilized. 6

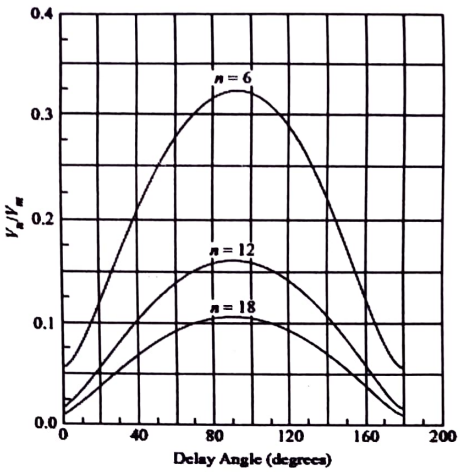


Fig. 3(d): Normalized output voltage harmonics as a function of delay angle for a three-phase rectifier.

PART B

Q4. a) Draw a typical circuit of a linear dc voltage regulator. What is the drawback of it? How can it be overcome? 6

b) Draw the circuit diagram and the waveform of inductor current, diode current and capacitor current for a typical buck converter. Also derive the relation between inductor current and diode current from the waveforms. 8

c) A boost converter has an input of 5V and an output of 25 W at 15 V. The minimum inductor current must be no less than 50% of the average. The ripple output must be less than 1 percent. The switching frequency is 300kHz. (i) Determine the duty ratio. (ii) 6
Minimum inductor value (iii) Minimum capacitor value 5

d) Differentiate between SEPIC converter and Ćuk converter.

Q5. a) Briefly explain the types of switch mode DC-DC converters used in a stand-alone solar PV system. 5

b) Draw the circuit of a three-level cascade multi source inverter. Why pattern swapping is important in multi-source inverter circuit? 6

c) Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150 V dc source. The load is a series RL combination with R=12Ω and L=60 mH. Select the switching frequency such that the current THD is less than 10%. Assume the impedance at the carrier frequency is entirely inductive. 10

Exp 8-9

Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	m_a=1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
n=1	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n=mf\pm2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

d) Determine the voltage THD of a square wave inverter. 4

Q6. a) A 10 Ohm resistive load requires a 1000-Hz, 50-V rms sinusoidal voltage. The THD of the load voltage must be no more than 5 percent. An adjustable dc source is available. (i) Design an inverter for this application. (ii) What is maximum voltage across the capacitor? 7

9-3

b) What is cyclo-converter? Mention its applications. 5

c) Explain the structure of TRIAC with diagram. Mention the constructional feature of a Power BJT. 8

d) Compare the following device in terms of power capacity and switching speed: 5

BJT, MOSFET, GTO, IGBT, MCT